

ITAI-1371-Intro to Machine Learning-RT-14274

Professor: Viswanatha Rao

Team ML_1371_14274 2

L04 Reflective Journal

Exploratory Data Analysis

This lab helped us understand that Exploratory Data Analysis is much more than simply looking at a dataset before building a machine learning model. We learned that EDA is an important part of the process because it allows us to identify missing values, unusual patterns, imbalances, relationships between variables, and possible problems that could affect the model later.

One of the first things that stood out to us in the Titanic dataset was the missing information. The dataset contains 891 passengers, but some columns do not have complete information. For example, Age and Cabin contain missing values, and Embarked also has a few missing entries. This helped us understand why data cleaning is so important. Before training a model, we need to decide how to handle missing values and whether some columns are useful enough to keep.

The visualizations were one of the most useful parts of the lab. Looking only at the overall survival rate gives us one piece of information, but the plots helped us see much more. We could clearly see differences in survival based on passenger class, gender, age, fare, and port of embarkation. This showed us how visualizations can reveal patterns that are difficult to notice by simply looking at numbers or averages.

The Fare analysis was also useful because it showed us how distributions can contain extreme values and outliers. The average fare does not fully represent the dataset because some passengers paid very high fares compared with most others. The boxplots helped us understand why it is important to look at the distribution and not rely only on the mean.

Another important lesson came from the Embarked analysis. We observed that passengers who embarked from different ports had different survival rates. However, we also realized that we should be careful when interpreting this type of pattern. A relationship between two variables does not automatically mean that one directly caused the other. Other variables, such as passenger class, may also influence the result. This helped us understand that EDA is useful for finding relationships, but those relationships still need to be interpreted carefully.

We also learned more about imbalanced data. In the Titanic dataset, more passengers did not survive than survived. This means the target classes are not evenly distributed. We now understand that this can become important later when evaluating a machine learning model because accuracy alone may not always give a complete picture of model performance.

One of our biggest takeaways from this lab is that EDA helps us ask better questions before we begin modeling. We need to understand what information we have, what is missing, what type of variables are included, whether there are outliers, how the target is distributed, and which features may have meaningful relationships with the outcome.

This lab also helped us understand the type of dataset we need for the Midterm project. We now see why a dataset that is already perfectly clean would give us fewer opportunities to demonstrate the complete machine learning workflow. A dataset with missing values, categorical and numerical variables, and other data quality issues allows us to practice exploration, cleaning, transformation, and preparation before modeling.

Overall, this lab made us more aware that the quality of a machine learning model depends heavily on understanding the data first. EDA gives us the opportunity to identify issues early, make informed decisions, and better understand the story behind the data before moving to model training.